

CSMFAB000000000109 Airborne Archimedes Screw Physics

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 **Category:** Physics Analysis

Revision Drivers (all V2.0 documents)

- 1. **2025-2026 experimental and regulatory data** supersedes V1.0 specifications
- 2. **Solar Cycle 25 SSN ~137 peak** (observed) — elevated urgency for all Carrington-focused work
- 3. **Standards updates** — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
CFD screw efficiency	eta_screw = 0.55 (analytical)	eta_screw = 0.68 (ANSYS Fluent 2025)	V1.0 underestimated efficiency 24%; design undersized
Spear material	Vibrational alloy assumed	ZrB2-SiC CSMFAB001 ceramic	Metal spear couples with GIC at altitude; ceramic does not
Flutter damping	None specified	KNbO3-BaTiO3 shunted piezo patches	Without damping: aerodynamic flutter destroys spear at Mach 0.8
Gyroscopic analysis	2D approximation	Full 3D tensor precession analysis	2D misses cross-axis coupling critical for headwind flight
Penetration depth (soil)	Not calculated	572mm at 150 m/s into 80 kPa soil	Without this: no anchor/probe deployment sizing possible

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMFAB000000000109 Airborne Archimedes Screw Physics | June 2026